

Python and Coding Agents (AI)

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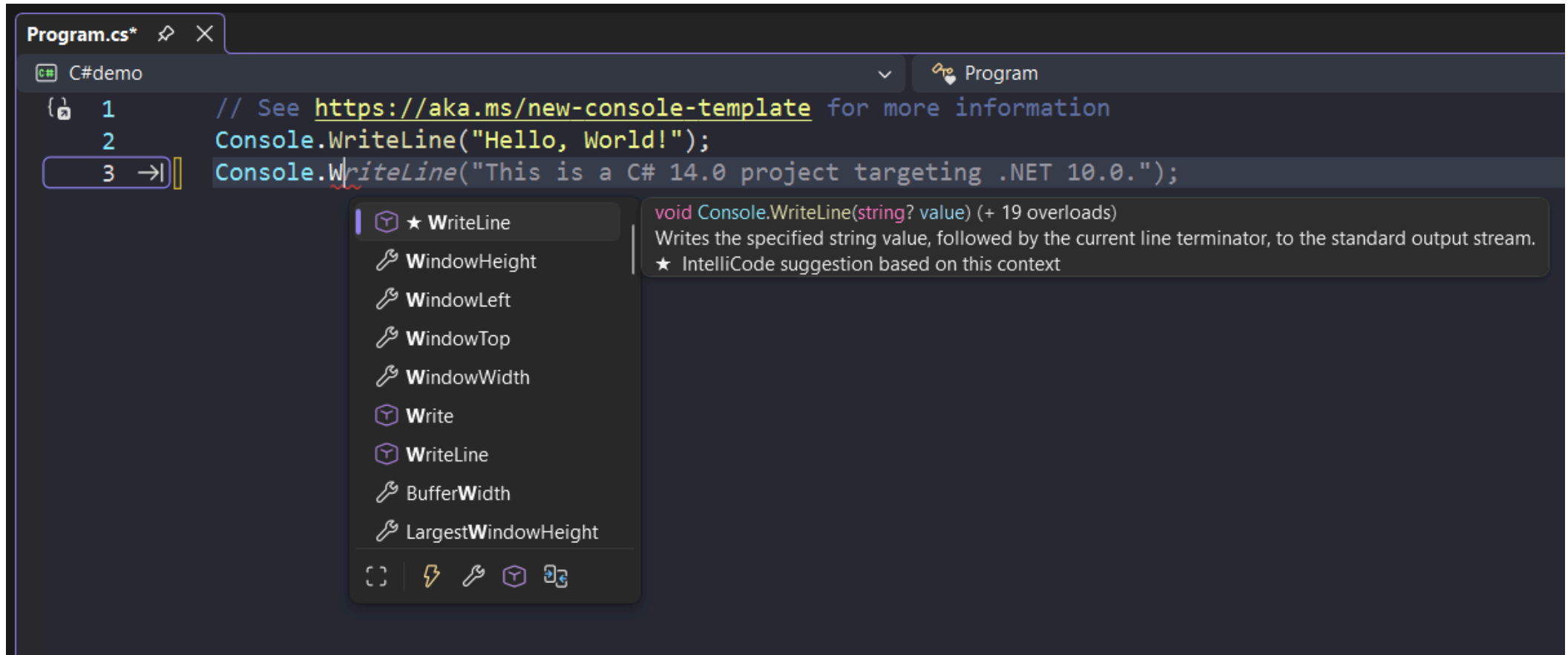
2026-06-10

Agenda

- Coding Agents Explained
- Hands-On Paper Replication

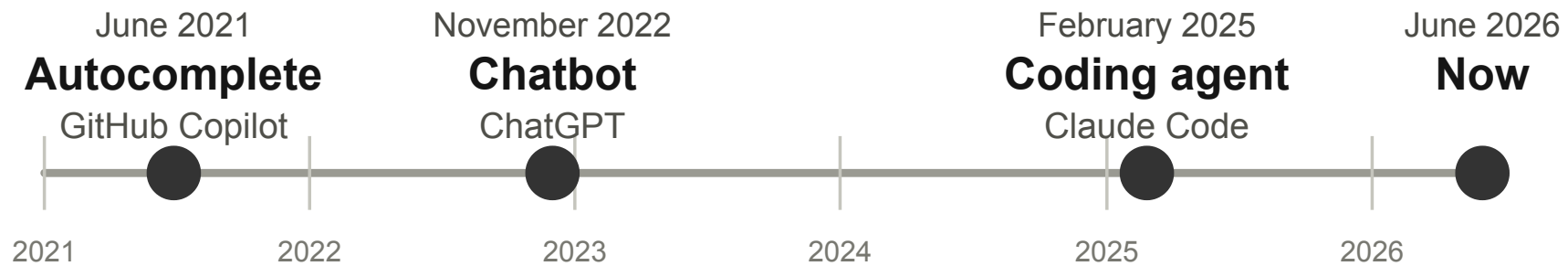
2021: Code Autocompletion

AI suggests the next code while you type. One can accept, ignore, or edit the suggestion. Github (Microsoft) released Copilot on June 29, 2021.



Visual Studio IntelliCode and Copilot autocomplete

AI Timeline



What is a Coding Agent?

It is AI that takes actions on your behalf (agent). We focus on the local coding agents (on your laptop).

- It can run commands and scripts
- It can read and write local files (code, papers, presentations, etc.)
- It can search the web, download datasets, etc.

Its like ChatGPT, but without having to copy-paste back and forth.

Coding Agent Example

presentation

python-and-agentic-coding.qmd

python-and-agentic-coding.qmd > Hands-On

```

38 # What is a Coding Agent?
47 Its like ChatGPT, but without having to copy-paste back and forth.
48
49 # Coding Agent Example
50
51 <video controls preload="metadata" width="100%">
52 | <source src="videos/agentic-example-transpose-table.mp4" type="video/mp4">
53 </video>
54
55 ## Largest Providers
56
57 | Company | ![OpenAI Logo](images/openai-logo.svg){width=150px} | ![Anthropic Logo](images/anthropic-logo.svg){width=150px} |
58 |---|---|---|
59 | Website | <https://chatgpt.com> | <https://claude.ai> |
60 | Chatbot | ChatGPT | Claude |
61 | ^ release date | Nov 30, 2022 | Mar 14, 2023 |
62 | Coding agent | Codex | Claude Code |
63 | ^ release date | Apr 16, 2025 | Feb 24, 2025 |
64
65 Coding agent is included in your 20€/month subscription. Pick whatever you like.
66
67 # Hands-On
68
69 Replication of AJR (2001), "The Colonial Origins of Comparative Development":
70 Economic performance is caused by institution, with colonial origins.
71
72 * Download dataset
73 * Plot figures
74 * Run regressions, create tables
75 * Compare to original results
76 * Write a report
77
78 # Takeaways
79
80 * Coding agents can speed up coding, researching, writing.
81 * They can make mistakes (!), we should check their results.
82 * Good judgement is more important than ever!

```

Press **Shift** **Tab** to automatically approve code edits

0:00 / 1:01

Agents Window

Python and Coding Agents

http://localhost:5877/python-a...

7 / 9

Largest Providers

Company	OpenAI	ANTHROPIC
Website	https://chatgpt.com	https://claude.ai
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Hands-On

Replication of AJR (2001), “The Colonial Origins of Comparative Development”: Economic performance is caused by institution, with colonial origins.

- Download dataset
- Plot figures
- Run regressions, create tables
- Compare to original results
- Write a report

Takeaways

- Coding agents can speed up coding, researching, writing.
- They can make mistakes (!), we should check their results.
- Good judgement is more important than ever!